Ivan Compagnucci, Ph.D.

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Academic Positions

Aug. 2024 – Now

PostDoctoral Fellow
Gran Sasso Science Institute, L'Aquila (AQ), Italy.
Supervisor: Prof. Catia Trubiani
Research Project: DREAM - PRIN 2022

Education

Nov. 2020 – Jul. 2024

Ph.D. in Computer Science & Mathematics
University of Camerino, Camerino (MC) Italy Thesis Title: Adaptive Digital Twin Supporting IoT-Enhanced Business Process.
Supervisor: Prof. Barbara Re.

Oct. 2017 – May 2020

M.Sc. Computer Science LM-18, University of Camerino
Thesis Title: Modeling and Enactment of Smart Environments combining Business Process and Internet of Things.
Mark: 110/110 cum laude.

Oct. 2014 – Jan. 2018

B.Sc. Computer Science L-31, University of Camerino
Thesis Title: Data Mining in the Management of a Non-Profit Organisation.
Mark: 110/110 cum laude.

Visiting Experiences

Mar. 2024 – Jul. 2024

Visiting Researcher (5 Months)
Research stay at the **Information Systems Engineering Research Group (LIRIS)** at **KU Leuven**, Brussels, under the supervision of Prof. Monique Snoeck and Prof. Estefania Serral Asensio. During the research stay, I worked on Digital Twins [11] and Learning Analytics in BPM [12].

Research Projects

- **DREAM - PRIN 2022.** Member of the founded national project **DREAM** modular software Design to Reduce uncertainty in Ethics-based cyber-physicAl systeMs. PNRR - PRIN 2022.
  Supervisor: Prof. Catia Trubiani
  Duration: 24 months

- **ALMONDO - PRIN 2022.** Member of the founded national project **ALMONDO** Develop a novel modeling framework to study climate lobbying that carefully considers the features of human learning and communication. PNRR - PRIN 2022.
  Supervisor: Dr. Fabrizio Fornari
  Duration: 12 months

Research Projects (continued)

- **FLUIDWARE - PRIN 2017.** Member of the founded national project **FLUIDWARE** Develop a novel programming model for IoT services and applications. MIUR - PRIN 2017.
  Supervisor: Prof. Barbara Re
  Duration: 36 months

Research Focus

Software Engineering for AI/GenAI

- Software architectures and performance evaluation for AI-based systems, with a particular focus on distributed learning systems. My work examines and evaluates how architectural design choices and runtime conditions influence both system-level properties, such as training time, communication efficiency, and resource usage, and learning-level properties, such as model accuracy. I also work on developing architectural and software engineering solutions, including architectural patterns, benchmarking frameworks, runtime adaptation mechanisms, and agentic solutions, to improve the efficiency, robustness, and overall quality of AI-based systems.

Business Process Management

- Empirical studies on business process modeling, with particular attention to the BPMN standard and its use in practice. My research focuses on large-scale analyses of BPMN models to investigate notation usage, practical complexity, element combinations, syntactic issues, and modeling practices. It also includes the development of tools for automated inspection and quantitative characterization of BPMN models.

IoT and Digital Twins

- Modeling and analysis of IoT-Enhanced Business Processes and Digital Twins, with particular attention to processes in which IoT devices, such as sensors and actuators, are integrated into process execution. My work focuses on conceptual architectures and model-based solutions for process monitoring, what-if analysis, and the alignment between physical devices and their digital representations through Digital Process Twins.

Keywords: AI-based Systems; AI for Software Engineering; Performance Evaluation; Software Architecture; Self-Adaptive Systems; Architectural Patterns; Distributed Learning; Federated Learning; Business Process Management; BPMN; Process Modeling; Process Analysis; Internet of Things; IoT-Enhanced Business Processes; Digital Twins; Digital Process Twins; Conceptual Modeling.

Research Activities

Organizing Committee

- **Co-Chair.** 2nd International Workshop on Software Architecture for Generative AI (SAGAI) at **ICSA26** [**Link**]

Program Committee

2026
- **PC-Member.** 21st International Conference on Software Engineering Advances (ICSEA) at **SoftNet'26** [**Link**]

Research Activities (continued)

- **PC-Member.** The 2026 IEEE International Conference on Digital Twin (DT'26) at **IEEE Smart World Congress 2026** [Link]
- **PC-Member.** 4th International Workshop on Business Processes Meet the Internet of Things (BP-Meet-IoT) at **CBI/EDOC'26** [Link]
- **PC-Member.** 4th International Workshop on Modelling and Implementation of Digital Twins for Complex Systems (MIDas4CS) at **CAiSE'26** [Link]
- **PC-Member.** 1st International Workshop on Engineering Autonomous Systems Intelligence (EASI) at **CAiSE'26** [Link]
- **PC-Member.** 1st International Workshop on AI Code QUality, Integrity & REliability (ACQUIRE) at **EDCC'26** [Link]

2025
- **PC-Member.** 1st International Workshop on Software Architecture for Data-Intensive Systems (SADIS) at **ECSA'25** [Link]

Review Activities

Elsevier
- **Journal of Systems and Software** [Link]
- **Future Generation Computer Systems** [Link]
- **Information and Software Technology** [Link]
- **Information Fusion** [Link]
- **Internet of Things** [Link]
- **Displays** [Link]
- **Science of Computer Programming** [Link]
- **SoftwareX** [Link]

Taylor & Francis
- **Cogent Business & Management** [Link]

Awards

2023
- 🏆 **BPM 2023 Best Paper Award (Demonstration and Resources Forum)**
Award received for the paper *BPMN Inspector: A Tool for Extracting Features from BPMN Models* [10], presented at the 21st International Conference on Business Process Management (BPM 2023). `https://bpm-conference.org/awards/`
The work was also highlighted in the November 2023 issue of the BPM Newsletter for its potential impact on BPM practitioners and educators. `https://bpm-conference.org/assets/docs/newsletter/BPM-newsletter-2023-11.pdf`

Teaching and Tutoring

2024
- **STEM, Digital and Innovation Skills – Teaching** (for Primary School)
Tutor in the PNRR project "Crescere per scoprire e aprirsi al mondo" (Mission 4 – Education and Research, Investment 3.1 "New Skills and New Languages", D.M. n. 65/2023), delivering educational and orientation activities aimed at strengthening STEM, digital, and innovation skills, with a focus on inclusion and equal opportunities.
Project code: M4C1I3.1-2023-1143-P-30633. C.U.P. G84D23005080006.

Teaching and Tutoring (continued)

2021 - 2023 **Operating Systems – Tutoring** (for M.Sc. Students)
I served as a tutor for the Operating Systems course at the University of Camerino during the academic years 2021/2022 and 2022/2023.

2020 - 2024 **Thesis Supervision**
During my Ph.D., I supervised 5+ bachelor's theses in the Computer Science program at the University of Camerino.

Third Mission Activities
I was involved in public engagement activities in schools, delivering lessons and workshops to students from kindergarten to upper secondary school. These activities focused on programming and computer science, with content adapted to different age groups. This activity reflects the universitys societal impact mission to support the goals of the Italian PNRR to promote STEM and digital skills across all levels of education.

Contributed Talks

2026 **[Speaker]** 21st International Conference on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS'26**) co-located with the International Conference on Software Engineering **ICSE'26**, Rio de Janeiro, Brazil, April 12–18, 2026.

2025 **[Speaker]** 4th Conference on System and Service Quality, (**QualITA'25**), Catania, Italy, June 25 and 27, 2025.

[Speaker] Experimenting Architectural Patterns in Federated Learning Systems
Seminar at Gran Sasso Science Institute, L'Aquila, 16 April 2025.

[Speaker] 22nd IEEE International Conference on Software Architecture (**ICSA '25**), Odense, Denmark, 31 March – April 4, 2025.

2024 **[Speaker] Trends on the use and adoption of the BPMN during the last ten years**
Seminar at the University of Trieste, Trieste, 18 December 2024.

[Speaker] [Session Chair] 28th International Conference on Enterprise Design, Operations, and Computing (**EDOC '24**), Vienna, Austria, September 10–13, 2024.

[Speaker] **Digital Process Twin: Open Challenges**
Seminar at Gran Sasso Science Institute, L'Aquila, 17 July 2024.

2023 **[Speaker]** 26th International Conference on Model-Driven Engineering Languages and Systems (**MODELS '23**), Västerås, Sweden, October 1–6, 2023.

[Speaker] 21st International Conference on Business Process Management (**BPM '23**), Utrecht, The Netherlands, September 11-15, 2023.

[Speaker] Trends on the adoption of BPMN during the last decade
Seminar at the University of Camerino, Camerino, 2023.

2022 **[Speaker] The use of the BPMN in IoT Environments**
Seminar at the University of Camerino, Camerino, 2022.

2021 **[Speaker]** 20th international Conference on Business Informatics Research (**BIR '21**), Vienna, Austria, September 22–24, 2021.

Contributed Talks (continued)

2020 **[Participant]** 18th International Conference on Business Process Management (**BPM '20**), Seville, Spain, September 13–18, 2020.

Research Publications

Journal Articles

J1 **Compagnucci, I.**, Pinciroli, R., & Trubiani, C. (2026). Experimenting Architectural Patterns in Federated Learning Systems. *Journal of Systems and Software (**JSS**)*, *232*, 112655. doi:10.1016/J.JSS.2025.112655

J2 Fornari, F., **Compagnucci I.**, Donato, M. C. D., Bertrand, Y., Beyel, H. H., Carrión, E., ... Valderas, P. (2025). Digital Twins of Business Processes: A Research Manifesto. ***Internet of Things***, *30*, 101477. doi:10.1016/J.IOT.2024.101477

J3 **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2024). A Study on the Usage of the BPMN Notation for Designing Process Collaboration, Choreography, and Conversation Models. *Business & Information Systems Engineering, (**BISE**)*, *66*, 43–66. doi:10.1007/S12599-023-00818-7

J4 **Compagnucci, I.**, Corradini, F., Fornari, F., Polini, A., Re, B., & Tiezzi, F. (2022). A Systematic Literature Review on IoT-Aware Business Process Modeling Views, Requirements and Notations. *Software and Systems Modeling, (**SoSyM**)*, *14*(1), 1–36. doi:10.1007/S10270-022-01049-2

Conference Proceedings

C1 Baresi, L., **Compagnucci, I.**, Lestingi, L., & Trubiani, C. (2026). Adaptive Toggling of Architectural Patterns for Federated Learning. In *International Conference on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**)*. To appear,

C2 Silvetti, S., **Compagnucci, I.**, Cairoli, F., Trubiani, C., & Nenzi, L. (2026). Time Robustness for Point-Based Semantics of Metric Interval Temporal Logic. In *Proceedings of the 23rd international conference on principles of knowledge representation and reasoning, (**KR**)*. To appear.

C3 **Compagnucci, I.**, Pinciroli, R., & Trubiani, C. (2025). Performance Analysis of Architectural Patterns for Federated Learning Systems. In *International Conference on Software Architecture, (**ICSA**)* (pp. 289–300). doi:10.1109/ICSA65012.2025.00036

C4 **Compagnucci, I.**, & Trubiani, C. (2025). Towards AI Agents for Selecting Architectural Patterns in Federated Learning Systems. In *4th Conference on System and Service Quality (QualITA 2025)*. (Vol. 4080).

C5 **Compagnucci I.**, Re, B., Asensio, E. S., & Snoeck, M. (2024). A Digital Process Twin Conceptual Architecture for What-If Process Analysis. In *Enterprise Design, Operations, and Computing. (EDOC) 2024 Workshops* (Vol. 537, pp. 373–388). doi:10.1007/978-3-031-79059-1_23

C6 **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2023). BPMN Inspector: A Tool for Extracting Features from BPMN Models. In *Proceedings of the Best Dissertation Award, Doctoral Consortium, and Demonstration & Resources Forum at (BPM)* (Vol. 3469, pp. 122–126).

C7 **Compagnucci, I.**, Snoeck, M., & Asensio, E. S. (2023). Supporting Digital Twins Systems Integrating the MERODE Approach. In *International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, MODELS-C* (pp. 449–458). doi:10.1109/MODELS-C59198.2023.00079

C8 Vemuri, P., Poelmans, S., **Compagnucci, I.**, & Snoeck, M. (2023). Using Formative Assessment and Feedback to Train Novice Modelers in Business Process Modeling. In *International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, MODELS-C* (pp. 449–458). doi:10.1109/MODELS-C59198.2023.00079

C9 **Compagnucci, I.**, Corradini, F., Fornari, F., & Re, B. (2021). Trends on the Usage of BPMN 2.0 from Publicly Available Repositories. In *International Conference on Business Informatics Research, (BIR)* (Vol. 430, pp. 84–99). doi:10.1007/978-3-030-87205-2_6

C10 **Compagnucci, I.**, Corradini, F., Fornari, F., Polini, A., Re, B., & Tiezzi, F. (2020). Modelling Notations for IoT-Aware Business Processes: A Systematic Literature Review. In *Business Process Management Workshops (BPM)* (Vol. 397, pp. 108–121). doi:10.1007/978-3-030-66498-5_9

Software

BPMN Inspector A Tool for Extracting Features from BPMN Models

Version. 1.0.0

BPMN Inspector for the analysis of collections of BPMN models. It supports research by automatically analyzing large collections of BPMN models and providing quantitative data on their characteristics. More specifically, it classifies models by type, detects duplicates and invalid models, and flags non-English models, thereby removing the need for manual inspection of these aspects. It also provides a detailed characterization of the analyzed collection by quantifying the use of BPMN notation elements and their combinations, identifying syntactic violations of the standard, and assessing if the models comply with established good modeling practices.

Link: https://github.com/PROSLab/BPMN-Inspector

Related References: [3, 10, 13]

BPM 2023 Best Paper Award (Demonstration and Resources Forum) [**Link**]

AP4Fed: A Federated Learning Benchmarking Platform

Version 1.5.0

AP4Fed is a Federated Learning platform built on top of Flower that supports the execution, monitoring, and adaptive configuration of Federated Learning processes. It extends Flower with a set of architectural patterns that can be dynamically enabled or disabled during training, allowing the system to adapt the execution strategy to the current conditions of the learning process. In particular, AP4Fed supports adaptive decisions on patterns such as client selection, heterogeneous data handling, and message compression, aiming to improve performance while introducing only limited overhead. It also provides monitoring capabilities that collect execution data and enable the evaluation of different adaptation strategies in experimental settings.

Link: https://github.com/IvanComp/AP4Fed

Related References: [5, 1, 7]

L'Aquila – April 30, 2026